Chapter 15

How to use CSS transitions, transforms, animations, and filters

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Objectives

Applied

1. Use any of the properties presented in this chapter for working with transitions, transforms, animations, and filters.

Knowledge

1. Describe the use of a CSS transition and explain how one can be used to create an accordion.
2. Describe the use of a CSS transform.
3. Describe the use of a CSS animation and explain how one can be used to create a slide show.
4. Describe the use of a CSS filter.

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Properties for working with transitions

```
transition
transition-property
transition-duration
transition-timing-function
transition-delay
```

The syntax for the shorthand transition property

```
transition: [property] [duration] [timing-function]
            [delay];
```


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A transition that occurs when the mouse hovers over a heading

Before the transition

Hover over this heading to see its transition

During the transition

Hover over this heading to see its transition

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The HTML for the element that will be transitioned

```
<h1>Hover over this heading to see its transition</h1>
```

The CSS for this two-property transition

```
h1 { font-size: 120%;
     color: blue;
     transition: font-size 2s ease-out 1s,
                 color 2s ease-in 1s; }
h1:hover {
    cursor: pointer;
    font-size: 180%;
    color: red; }
```


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A transition for one property

```
h1 { font-size: 120%;
     transition: font-size 2s ease-out; }
h1:hover { font-size: 180%; }
```

A transition for all properties

```
transition: all 2s ease-out 1s
```


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An accordion created with CSS transitions

jQuery FAQs

What is jQuery?

Why is jQuery becoming so popular?

Three reasons:

- It's free.
- It lets you get more done in less time.
- All of its functions are cross-browser compatible.

Which is harder to learn: jQuery or JavaScript?

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The HTML for the accordion

```
<h1>jQuery FAQs</h1>
<div id="accordion">
    <h3><a href="#Q1">What is jQuery?</a></h3>
    <div id="Q1">Contents for first panel</div>
    <h3><a href="#Q2">Why is jQuery becoming so
                      popular?</a></h3>
    <div id="Q2">Contents for second panel</div>
    <h3><a href="#Q3">Which is harder to learn:
                      ...?</a></h3>
    <div id="Q3">Contents for third panel</div>
</div>
```

The CSS for the transitions

```
#accordion div {
    overflow: hidden;
    height: 0;
    transition: height 2s ease-in-out;
}
#accordion div:target { height: 120px; }
```


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Properties for working with 2D transforms

```
transform, transform-origin
```

Methods for 2D transforms

```
rotate(angle)
rotateX(angle)
rotateY(angle)
scaleX(value)
scaleY(value)
scale(x-value,y-value)
skewX(angle)
skewY(angle)
skew(x-angle,y-angle)
translateX(value)
translateY(value)
translate(x-value,y-value)
matrix(a,b,c,d,e,f)
```


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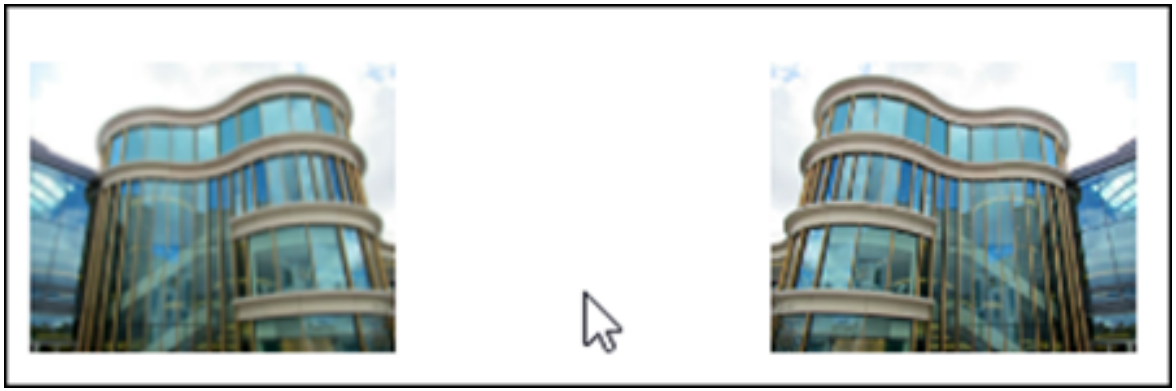
An image with a right origin that's rotated right



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The HTML for the original and rotated images

```
<p><img src="images/01.jpg" >
   <img src="images/01.jpg" class="image1">
</p>
```

The CSS for the transform

```
.image1 {
    transition: 2s; }
.image1:hover {
    transform: rotateY(180deg);
    transform-origin: right; }
```


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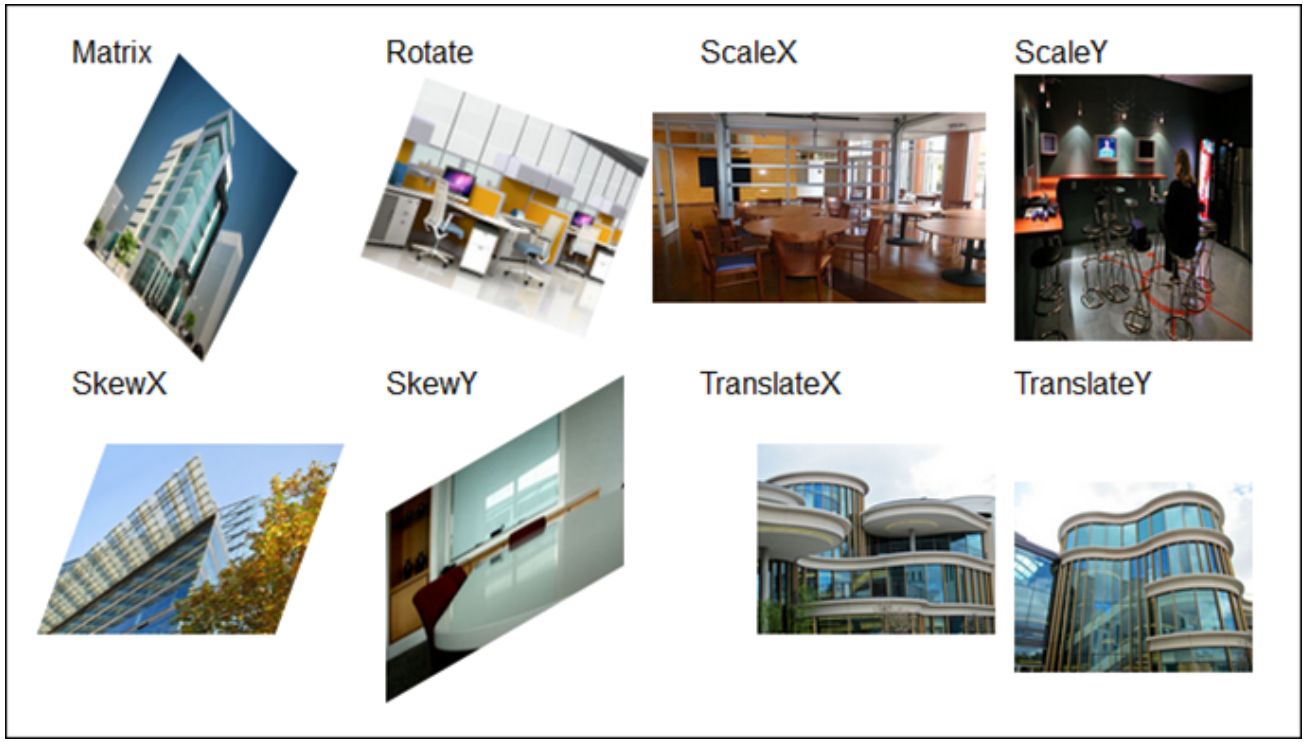
Images with 8 different transforms applied



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The HTML for the images with transforms

```
<ul>
    <li><p>Matrix</p>
        <img src="images/01.jpg" class="image1"></li>
    <li><p>Rotate</p>
        <img src="images/02.jpg" class="image2"></li>
    <li><p>ScaleX</p>
        <img src="images/03.jpg" class="image3"></li>
    <li><p>ScaleY</p>
        <img src="images/04.jpg" class="image4"></li>
    <li><p>SkewX</p>
        <img src="images/05.jpg" class="image5"></li>
    <li><p>SkewY</p>
        <img src="images/06.jpg" class="image6"></li>
    <li><p>TranslateX</p>
        <img src="images/07.jpg" class="image7"></li>
    <li><p>TranslateY</p>
        <img src="images/08.jpg" class="image8"></li>
</ul>
```


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The CSS for the 2D transforms

```
.image1 {transform: matrix(0.5,0.5,-0.5,1,0,0);}
.image2 {transform: rotate(20deg);}
.image3 {transform: scaleX(1.4);}
.image4 {transform: scaleY(1.4);}
.image5 {transform: skewX(-20deg);}
.image6 {transform: skewY(-30deg);}
.image7 {transform: translateX(30px);}
.image8 {transform: translateY(20px);}
```


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Properties for working with animations

```
animation
animation-name
animation-duration
animation-delay
animation-iteration-count
animation-timing-function
animation-direction
```

The syntax for the shorthand animation property

```
animation: [name] [duration] [timing-function] [delay]
           [iteration-count] [direction];
```


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A simple animation that moves a heading and changes its color

Starting in blue with a left margin of 20%

This text will animate.

Ending in red with a left margin of 60%

This text will animate.

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The HTML for the heading

```
<h1>This text will animate.</h1>
```

The CSS for the animation

```
h1 {animation: moveright 3s 2s infinite ease-in-out
               alternate; }
@keyframes moveright {
    from { margin-left: 20%;
           color: blue; }
    to {   margin-left: 60%;
           color: red; }
}
```


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A slide show animation with captions

Vecta Corp HQ

Rear of Building

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The HTML for an unordered list that contains the images and captions

```
<ul>
    <li>
        <h2>Front of Building</h2>
        <img src="images/01.jpg" alt="">
    </li>
    ...
</ul>
```

The CSS for the ul and li elements

```
ul { list-style: none;
     width: 500%;
     position: relative; }
ul li { width: 20%;
        float: left; }
```


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The CSS for the animation

```
ul { animation: slideshow 15s infinite alternate; }
@keyframes slideshow {
    0%    {left:    0%;}
    10%   {left:    0%;}
    20%   {left: -100%;}
    30%   {left: -100%;}
    40%   {left: -200%;}
    50%   {left: -200%;}
    60%   {left: -300%;}
    70%   {left: -300%;}
    80%   {left: -400%;}
    90%   {left: -400%;}
    100%  {left: -400%;}
}
```


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The syntax for adding a filter to an element

```
filter: filtermethod(value);
```

The filter methods

```
blur(value)
brightness(value)
contrast(value)
drop-shadow(values)
grayscale(value)
hue-rotate(angle)
invert(value)
opacity(value)
saturate(value)
sepia(value)
```


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An image before and after its colors are inverted

The HTML for the images

```
<li>Original<br><img src="images/01.jpg" alt=""></li>
<li>Inverted<br><img src="images/01.jpg" class="image1"
                     alt=""></li>
```

The CSS for the filter

```
.image1:hover { filter: invert(.80); }
```


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Different filters applied to an image in Chrome

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The HTML for the images

```
<ul>
  <li>Blur<br>        <img src="images/01.jpg" class="image1"
                           alt=""></li>
  <li>Brightness<br>  <img src="images/01.jpg" class="image2"
                           alt=""></li>
  <li>Contrast<br>    <img src="images/01.jpg" class="image3"
                           alt=""></li>
  <li>Drop Shadow<br> <img src="images/01.jpg" class="image4"
                           alt=""></li>
  <li>Grayscale<br>   <img src="images/01.jpg" class="image5"
                           alt=""></li>
  <li>Hue Rotate<br>  <img src="images/01.jpg" class="image6"
                           alt=""></li>
  <li>Invert<br>      <img src="images/01.jpg" class="image7"
                           alt=""></li>
  <li>Opacity<br>     <img src="images/01.jpg" class="image8"
                           alt=""></li>
  <li>Saturate<br>    <img src="images/01.jpg" class="image9"
                           alt=""></li>
  <li>Sepia/blur<br>  <img src="images/01.jpg" class="image10"
                           alt=""></li>
</ul>
```


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The CSS for the filters

```
.image1 { filter: blur(2px); }
.image2 { filter: brightness(50%); }
.image3 { filter: contrast(50%);}
.image4 { filter: drop-shadow(2px 2px 5px #333); }
.image5 { filter: grayscale(50%); }
.image6 { filter: hue-rotate(90deg); }
.image7 { filter: invert(.8); }
.image8 { filter: opacity(.50); }
.image9 { filter: saturate(30%); }
.image10{ filter: sepia(100%) blur(1px); }
```


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Short 15-1 Use a transition and transform

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Short 15-2 Use an animation

San Joaquin Valley Town Hall

Celebrating our 75th Year

Guest speakers

October
David Brancaccio

November
Andrew Ross Sorkin

January
Amy Chua

February
Scott Sampson

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Fossil Threads in the Web of Life

February
Scott Sampson

What's 75 million years old and brand spanking new? A teenage Utahceratops! Come to the Saroyan, armed with your best dinosaur roar, when Scott Sampson, Research Curator at the Utah Museum of Natural History, steps to the podium. Sampson's research has focused on the ecology and evolution of late Cretaceous dinosaurs and he has conducted fieldwork in a number of countries in Africa.

Scott Sampson is a Canadian-born paleontologist who received his Ph.D. in zoology from the University of Toronto. His doctoral work focused on two new species of ceratopsids (horned dinosaurs) from the Late Cretaceous of Montana, as well as the growth and function of certopsid horns and frills.

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